AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2015

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

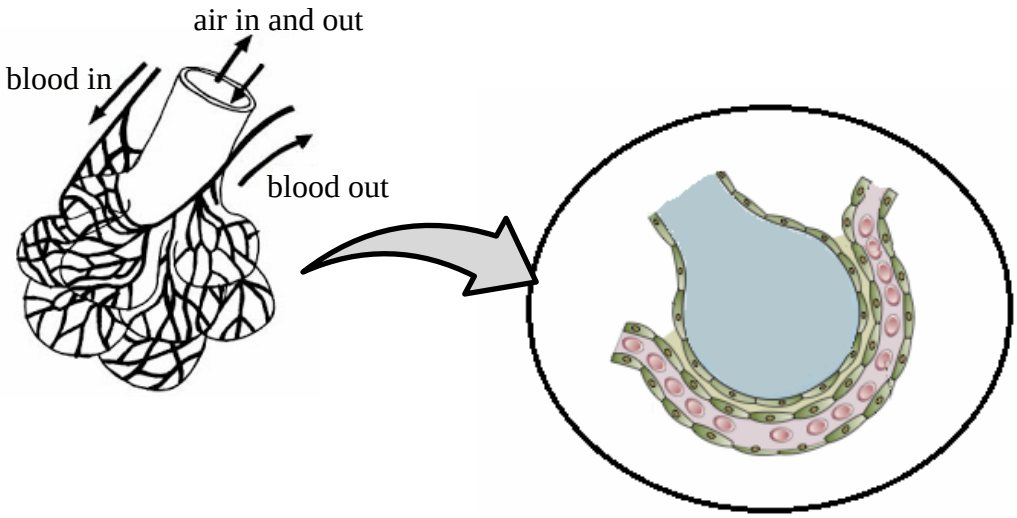
3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. The diagram below shows some characteristics of alveoli.



Which TWO characteristics, visible in the diagram, enable alveoli to perform their function efficiently? (2 Marks)

_____

_____

_____

_____

b. How do kidneys keep the composition of blood constant? (2 Marks)

_____

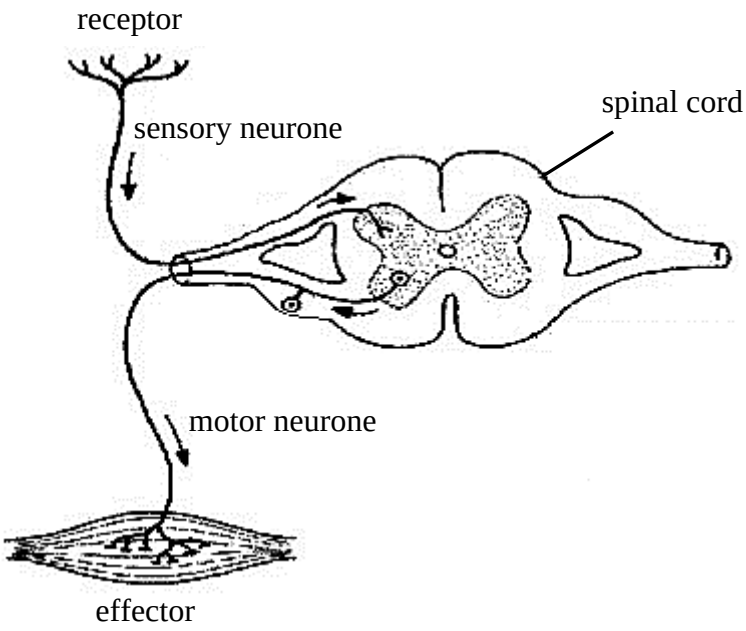
_____

_____

_____

Q.2. (Total 4 Marks)

The diagram shows the path of a nervous impulse in case of a reflex action.



There are TWO errors (mistakes) in the diagram. Highlight these errors (mistakes) in the diagram using circles and redraw the correct diagram.

Space for drawing

Q.3. (Total 4 Marks)

a. Categorize the following parts of the human skeleton as axial or appendicular. (2 Marks)

rib cage	**skull**	**pectoral girdle**	**vertebral column**

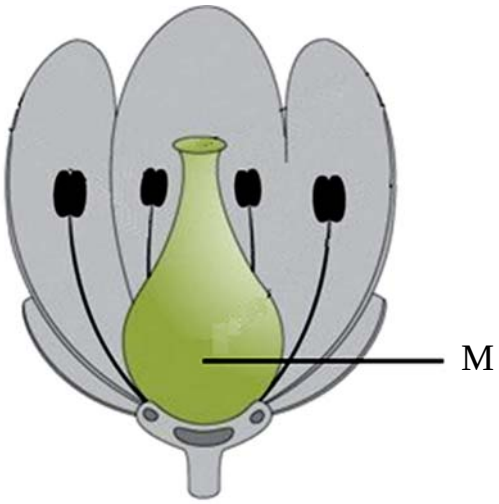
Axial Skeleton	Appendicular Skeleton

b. How do hinge joints work? Give an example of a hinge joint present in the human body. (2 Marks)

Q.4. (Total 4 Marks)

a. Why is parthenogenesis considered a type of asexual reproduction? (2 Marks)

b. The given diagram shows a part of a flower labelled M. What role does it play in the formation of a fruit? (2 Marks)



Q.5. (Total 4 Marks)

A male black-haired cat is crossed with a female white-haired cat. Determine the expected genotypes of the offspring.

(Your answer should include working of both the possible crosses.)

Cross 1	Cross 2
Expected Genotype:	Expected Genotype:

Q.6. (Total 5 Marks)

a. In what ways genetic engineering has benefited agriculture? (2 Marks)

i. _____

_____

ii. _____

_____

b. Three persons Mr X, Y and Z are addicted to sedatives, narcotics and hallucinogens respectively. Write any TWO effects that can be observed on the metabolism of each individual.

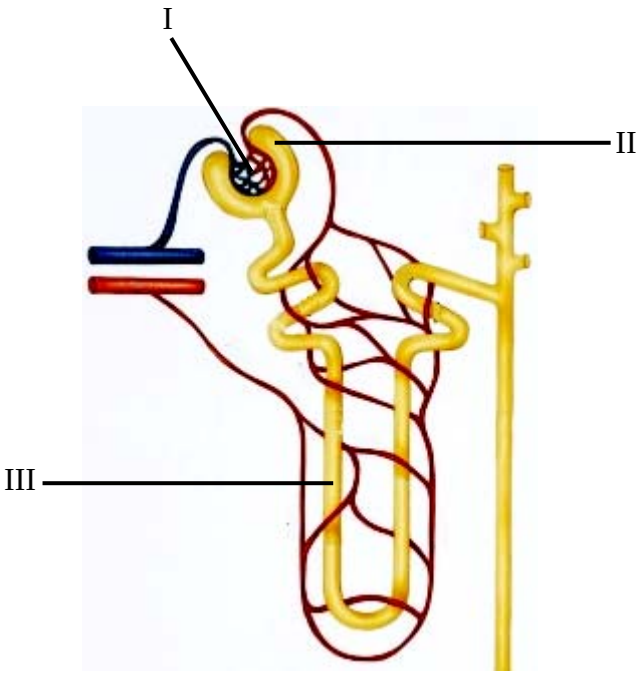
(3 Marks)

Person	Addicted to	Effect on Metabolism
Mr X	Sedatives	
Mr Y	Narcotics	
Mr Z	Hallucinogens	

Q.7. (Total 7 Marks)

EITHER

a. What is the significance of structural adaptation to function? Describe how the structure of each labelled part of the nephron is related to its respective function.



OR

b. Define the term hormone. Describe the effects of vasopressin, calcitonin and thyroxin on the body metabolism when released in excess and in less than required amount.

Q.8. (Total 8 Marks)

EITHER

a. Why is sun considered to be the principle source of energy for all forms of life? Describe the ways in which the flow of material and the flow of energy in an ecosystem are similar and different.

OR

b. Describe any TWO harmful effects of each, deforestation and acid rain, on plant and animal life.

END OF PAPER

Please use this page for rough work

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